Question 1

What is the next number in the following series of numbers?

-3, -8, -1, -6, 1, -4, ?

A) 2

B) 3

C) 4

D) 5

WORKSPACE

Question 2

What is the next number in the following series of numbers?

1/8, 1/14, 1/12, 1/21, 1/16, 1/28, ?

A) 1/18

B) 1/19

C) 1/20

D) 1/22

WORKSPACE

Question 3

What is the next number in the following series of numbers?

1/3, 1/8, 1/11, 1/19, 1/30, 1/49, ?

A) 1/68

B) 1/79

C) 1/81

D) 1/98

WORKSPACE

Question 4

What is the next number in the following series of numbers?

128, 109, 93, 80, 70, ?

A) 61

B) 62

C) 63

D) 64

WORKSPACE

Question 5

What is the next number in the following series of numbers?

1, 4, 13, 40, 121, ?

A) 362

B) 363

C) 364

D) 365

WORKSPACE

Question 6

What is the next number in the following series of numbers?

8, 23, 67, 198, 590, ?

A) 1768

B) 1769

C) 1765

D) 1770

WORKSPACE

Question 7

Simplify the following expression:

$$-2(3x^3 - (1/2)x^2 + 4x - 5) + x^3 - (3/2)x^2 - 6x + 7$$

A) $-5x^3 - (1/2)x^2 - 14x + 17$

B) $-5x^3 - (5/2)x^2 - 14x + 17$

C) $-5x^3 + (5/2)x^2 - 14x + 17$

D) $-7x^3 - (1/2)x^2 - 14x + 17$

WORKSPACE

Question 8

Consider the following formula:

$$y = -2(x - 3)(x + 5)$$

A) $-2x^2 - 4x + 30$

B) $-2x^2 - 10x + 30$

C) $2x^2 + 4x - 30$

D) $-2x^2 - 4x + 30$

WORKSPACE

Question 9

Factor the polynomial:

$$18x^4y^3 + 27x^3y^2$$

A) $9x^3y^3(2x + 3)$

B) $18x^3y^2(xy + 1)$

C) $9x^3y^2(2xy + 3)$

D) $9x^2y^2(2x^2y + 3x)$

WORKSPACE

Question 10

Factor the polynomial:

$6x^2 + 11x - 10$

A) (3x + 2)(2x - 5)

B) (6x - 2)(x + 5)

C) (3x + 5)(2x - 2)

D) (3x - 2)(2x + 5)

WORKSPACE

Question 11

Consider the formula. If B = 1 and C = 6, what is the value of A?

$$2(2A - (B + (1/2)C)) + 5 = A + C$$

A) 1

B) 2

C) 3

D) 4

WORKSPACE

Question 12

Evaluate the expression for N = 6 and P = (2N + 6)/3.

(N + 6)/P + P/2 = ?

A) 4

B) 5

C) 6

D) 8

WORKSPACE

Question 13

Consider the formula. If A = 4 and D = 5, what is the value of B?

$$D(B + 2) - A((1/2)B - 3) = 30 - (2B - A(B - 1))$$

A) 2

B) 3

C) 4

D) 5

WORKSPACE

Question 14

Consider the formula. If w = 5, x = 3, and y = 1, what is the value of z?

$$[3(2w - (x - y)^2) + 3(x + y)^2]/3 - z = 2w + 2xy$$

A) 6

B) 7

C) 8

D) 9

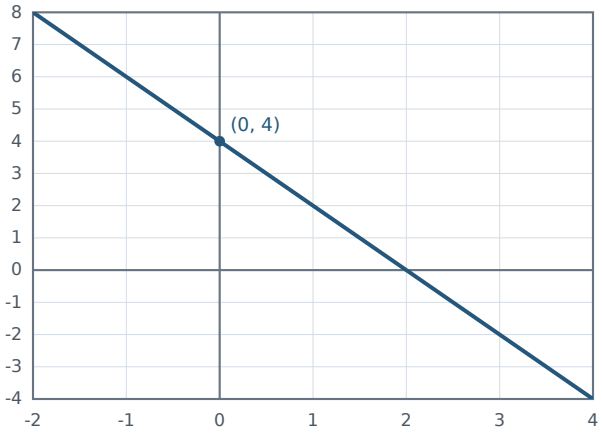
WORKSPACE

Question 15

Which choice represents the same relationship as the given function?

$y = -3x + 4$

A)

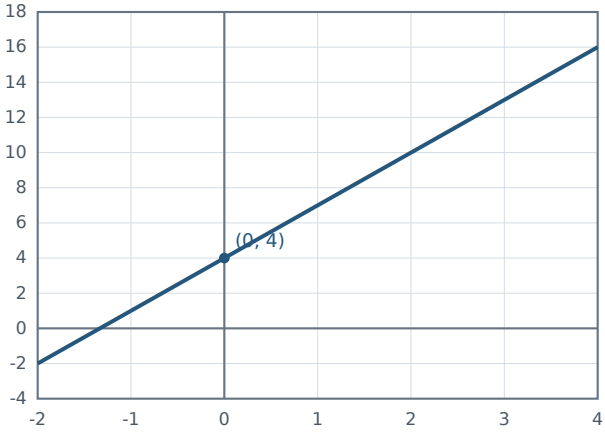


B)

x	y
0	4
1	2
2	0
3	-2
4	-4

C) The value of y is equal to 4 minus 3 times x.

D)



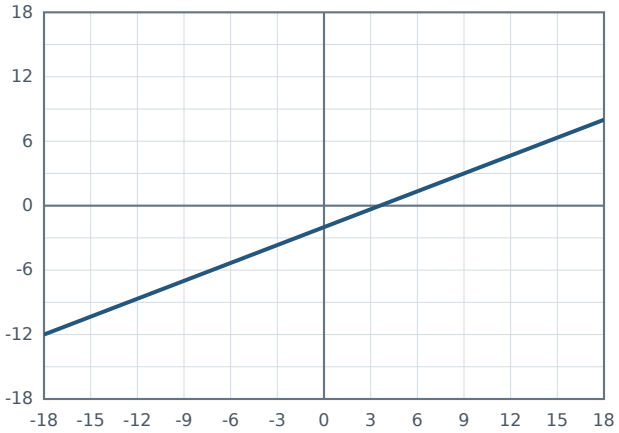
WORKSPACE

Question 16

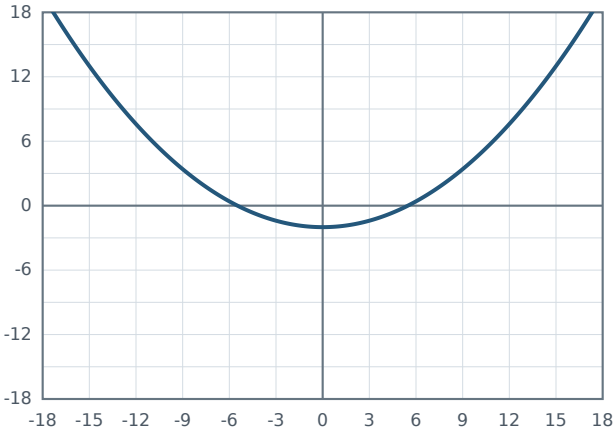
Which choice represents the same relationship as the information below?

y is equal to five-ninths of x, plus 2.

A)



B)



C)

x	y
0	2
9	6
18	10
27	14
36	18

D)

x	y
-36	-18
-27	-13
-18	-8
-9	-3
9	7

WORKSPACE

Question 17

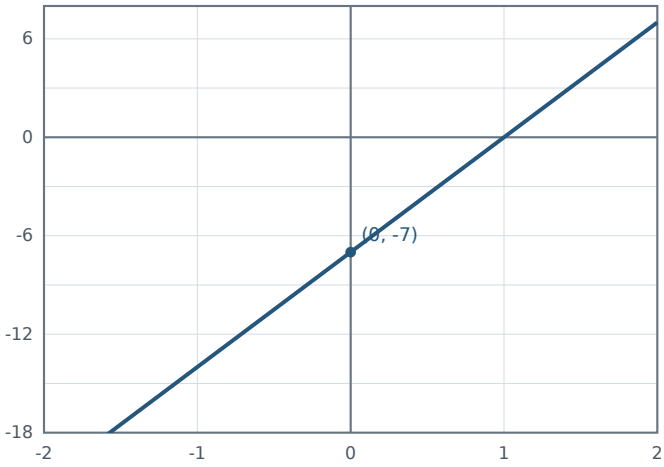
Which choice represents the same relationship as the given table?

C	T
-2	-21
-1	-14
0	-7
1	0
2	7

A) T is equal to 7 less than C.

B) T is equal to 7 times the quantity C - 7.

C)



D) T = 6(C - 1/2) + C - 3

WORKSPACE

Question 18

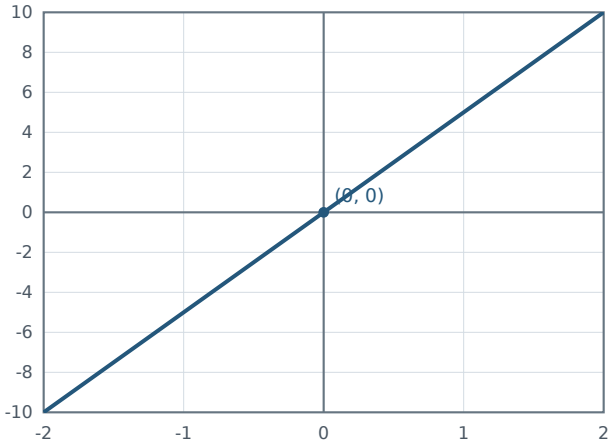
Simplify the expression and determine which relationship it represents:

$$y = [14x + 6(2x - (y - x)) + 4(y - 3x)]/2$$

A)

x	y
0	0
1	4
2	8
3	12
4	16

B)



C) The value of y is equal to the sum of x and four times the value of x, increased by 1.

D) None of the above.

WORKSPACE

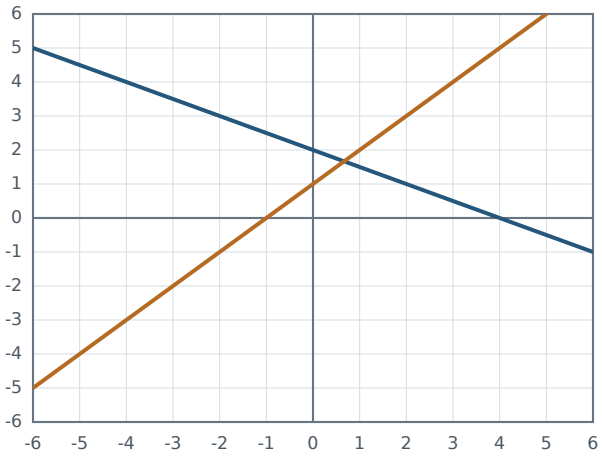
Question 19

Which graph represents the following two linear equations?

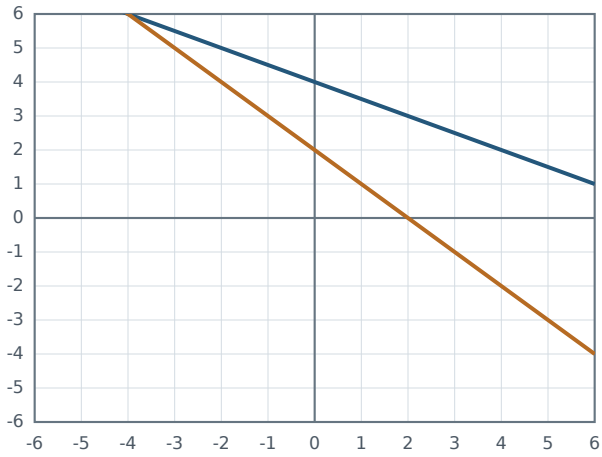
$y = -(1/2)x + 3$

$x + y = 1$

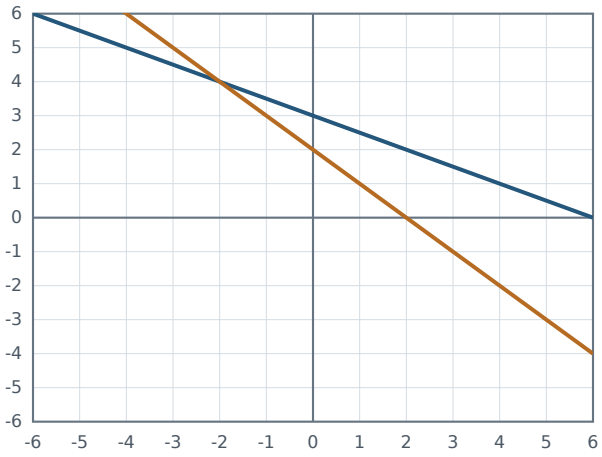
A)



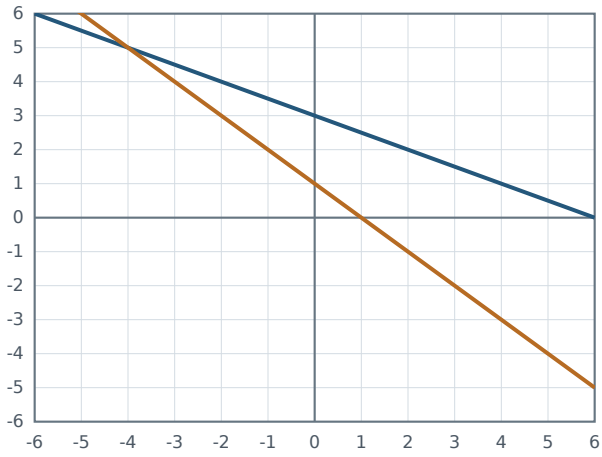
B)



C)



D)

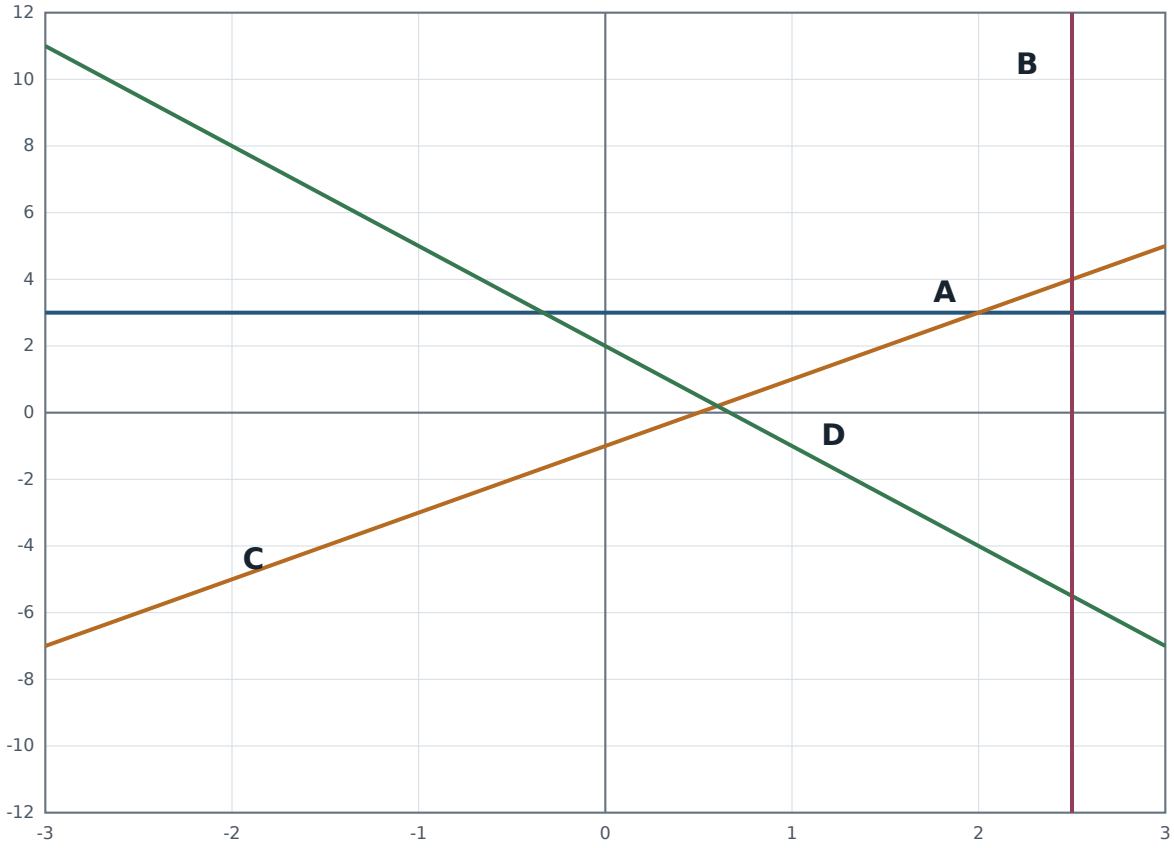


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Question 20

Which of the graphs represents the function?

$y = 3$

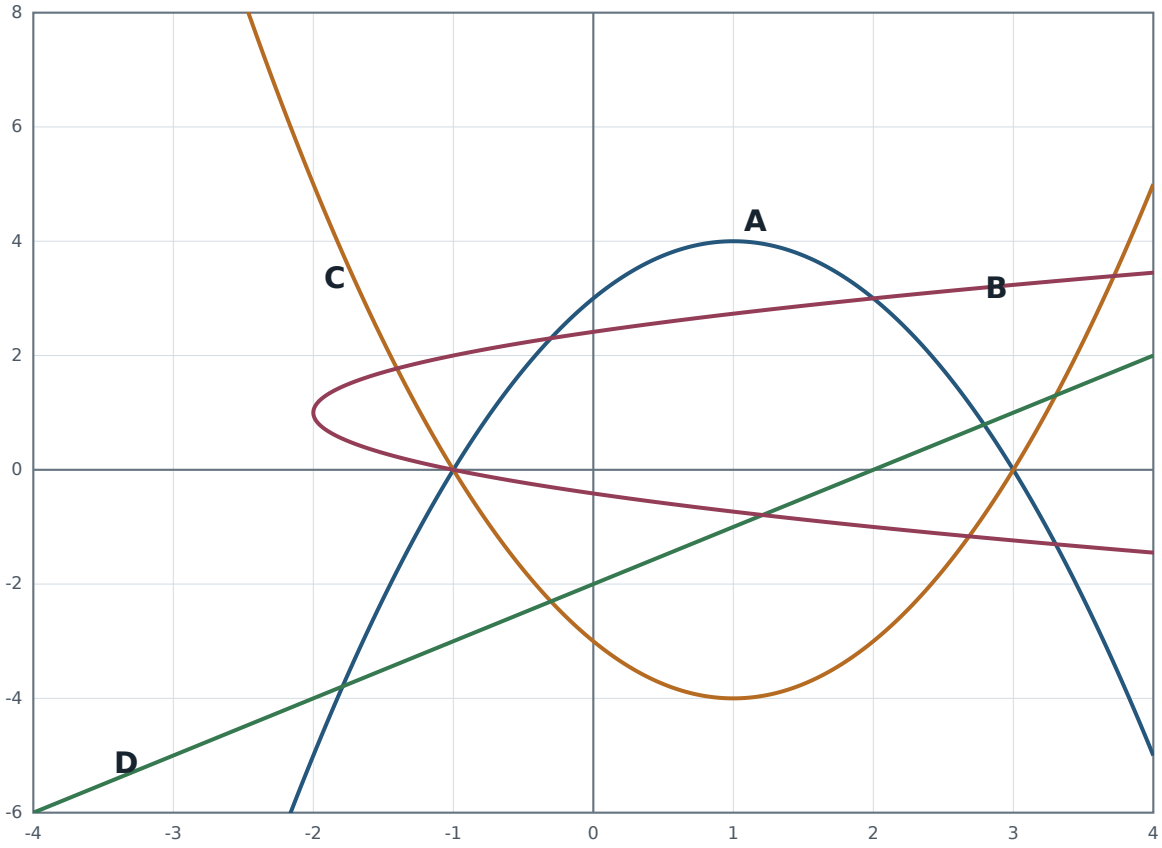


WORKSPACE

Question 21

Which of the graphs represents the function?

$y = -(x - 1)^2 + 4$



WORKSPACE

Question 22

Consider the formula. Where y > 2 and is held constant, which statement is true?

$$0 = 6x/(y - 2) - z$$

A) If x is negative, then z is positive.

B) If x increases, then z decreases.

C) If x is positive, then z is positive.

D) If z is positive, then x is negative.

WORKSPACE

Question 23

Consider the formula. Which statement is true?

$$x = 0.5y - 6$$

A) When the value of y is greater than 12, x is positive.

B) When the value of y is less than 12, x is positive.

C) When the value of y is greater than 12, x is negative.

D) When the value of y is less than 12, x is zero.

WORKSPACE

Question 24

Consider the formula. Which statement is true?

$$Z = (3/2)R - 4$$

A) The difference between the values Z and R is always 4.

B) If R is greater than 8/3, then Z is positive.

C) For positive Z values, $R < 0$.

D) For negative R values, $Z > 0$.

WORKSPACE

Question 25

Consider the formula. Which statement is true?

$$x = 12 - 2y$$

A) When the value of y is greater than 6, x is positive.

B) When the value of y is less than 6, x is positive.

C) When the value of x is less than 12, y is negative.

D) When the value of x is greater than 12, y is positive.

WORKSPACE

Question 26

Consider the formula. Which statement is not true?

$$3K = 6M + 18$$

A) When the value of M is greater than -3, K is positive.

B) When the value of M is equal to -3, K is zero.

C) When the value of M is less than -3, K is negative.

D) When the value of M is less than -3, K is positive.

WORKSPACE

Question 27

Consider the formula. Which statement is true?

$$Z = 2R - 6$$

A) $Z + 6 > 2R$

B) $2R - Z < 0$

C) $R - Z > R$

D) $(2R - Z) > (Z - 2R)$

WORKSPACE

Question 28

Solve the proportion. Which value of K is correct?

$$(K - 8)/[-(4 - K)] = 3/9$$

A) 5

B) 7

C) 10

D) 14

WORKSPACE

Question 29

Evaluate the formula:

$$Q = 3/r$$

$$r = (n + 2n)/(4n)$$

$$Q = ?$$

A) 4n/n

B) 4n/(3n)

C) 3n/(4n)

D) 12n/(3n)

WORKSPACE

Question 30

Solve the system of equations. Which ordered pair (a, b) is the solution?

$$2a + 5b = 11$$

$$3a - 6b = 3$$

A) (2, 1)

B) (3, 1)

C) (3, 2)

D) (4, 1)

WORKSPACE

Question 31

Find the unknown variable B, given I = 18, T = 10, C = 4, H = 2.

$$I/(T - C) + I(C - H)/[T - (H - 1)] - I/(C + H) = B$$

A) 1

B) 3

C) 4

D) 7

WORKSPACE

Question 32

Isolate C from the formula:

$$P = (Q - R)/S + TC - U$$

$$C = ?$$

A) [P - (Q - R)/S + U]/T

B) [P - (Q - R)/S - U]/T

C) T[P - (Q - R)/S + U]

D) (Q - R)/S + TC - P - U

WORKSPACE

Question 33

Isolate k_1 from the formula:

$$m_1 = [k_1(k_2 - m_2) + k_1m_2]/(m_2 + 2)$$

$$k_1 = ?$$

A) $m_1(m_2 + 2)/k_2$

B) $k_2/[m_1(m_2 + 2)]$

C) $m_1(k_2 + 2)/m_2$

D) $m_1(m_2 - 2)/k_2$

WORKSPACE

Question 34

Solve for x. Which is the correct solution?

$$13x^2 - 65x + 78 = 0$$

A) x = 2, x = 3

B) x = 3, x = 4

C) x = 1, x = 6

D) x = -2, x = -3

WORKSPACE